Transfer Function of 3rd-Order System: $H(s) = H_1(s) \cdot H_2(s)$

$$H_1(s) = \frac{1}{1 + a_1 \cdot s} \qquad H_2(s) = \frac{b_0}{1 + a_2 \cdot s + a_3 \cdot s^2}$$

--> a1, a2 and a3 are the filter coefficients from the tables in the Tietze-Schenk book.

$$(1 + a_1 \cdot s) \cdot (1 + a_2 \cdot s + a_3 \cdot s^2) \xrightarrow{expand} a_1 \cdot a_3 \cdot s^3 + a_3 \cdot s^2 + a_1 \cdot a_2 \cdot s^2 + a_2 \cdot s + a_1 \cdot s + 1$$

$$\alpha_1 = a_1 + a_2 \qquad \alpha_2 = a_1 \cdot a_2 + a_3 \qquad \alpha_3 = a_1 \cdot a_3 \qquad \beta_0 = b_0$$

$$H(s) = \frac{\beta_0}{1 + \alpha_1 \cdot s + \alpha_2 \cdot s^2 + \alpha_3 \cdot s^3} = \left(1 + \frac{s}{\omega_p}\right) \cdot \left(1 + \frac{s}{\omega_0 \cdot Q} + \left(\frac{s}{\omega_0}\right)^2\right)$$

$$\left(1 + \frac{s}{\omega_p}\right) \cdot \left(1 + \frac{s}{\omega_0 \cdot Q} + \left(\frac{s}{\omega_0}\right)^2\right) \xrightarrow{expand} \frac{s^3}{{\omega_0}^2 \cdot \omega_p} + \frac{s^2}{{\omega_0}^2} + \frac{s^2}{\omega_0 \cdot \omega_p \cdot Q} + \frac{s}{\omega_p} + \frac{s}{\omega_0 \cdot Q} + 1$$

- Coefficient Comparison:

$$\alpha_1 = \frac{1}{\omega_p} + \frac{1}{\omega_0 \cdot Q} \quad \sim \quad \alpha_1 = \frac{1}{\omega_p} \quad , \ \omega_0 \cdot Q > \omega_p$$

$$\alpha_2 = \frac{1}{{\omega_0}^2} + \frac{1}{\omega_0 \cdot \omega_p \cdot Q}$$

$$\alpha_3 = \frac{1}{{\omega_0}^2 \cdot \omega_p}$$

- Rearranging:

$$\omega_p = \frac{1}{\alpha_1} \cdot \omega_c = \frac{1}{a_1 + a_2} \cdot \omega_c$$

$$\omega_0 = \sqrt{\frac{\alpha_1}{\alpha_3}} \cdot \omega_c = \sqrt{\frac{a_1 + a_2}{a_1 \cdot a_3}} \cdot \omega_c$$

$$Q = \frac{\alpha_1 \cdot \alpha_3 \cdot \sqrt{\frac{\alpha_1}{\alpha_3}}}{\alpha_1 \cdot \alpha_2 - \alpha_3} = \frac{(a_1 + a_2) \cdot a_1 \cdot a_3 \cdot \sqrt{\frac{a_1 + a_2}{a_1 \cdot a_3}}}{(a_1 + a_2) \cdot (a_1 \cdot a_2 + a_3) - a_1 \cdot a_3} = \frac{(a_1 + a_2) \cdot a_1 \cdot a_3 \cdot \sqrt{\frac{a_1 + a_2}{a_1 \cdot a_3}}}{{a_1}^2 \cdot a_2 + a_2 \cdot a_3 + a_1 \cdot {a_2}^2}$$

3rd-Order Filter Coefficients (Tietze Schenk Table):

$a_1 := 1.0000$ $a_2 := 1.0000$ $a_3 := 1.0000$

$$\omega_p := \frac{1}{a_1 + a_2} \cdot \omega_c \rightarrow 0.5 \cdot \omega_c$$

$$\omega_0 := \sqrt{\frac{a_1 + a_2}{a_1 \cdot a_3}} \cdot \omega_c \rightarrow 1.4142135623730950488 \cdot \omega_c$$

$$Q := \frac{(a_1 + a_2) \cdot a_1 \cdot a_3 \cdot \sqrt{\frac{a_1 + a_2}{a_1 \cdot a_3}}}{{a_1}^2 \cdot a_2 + a_2 \cdot a_3 + a_1 \cdot {a_2}^2} = 0.943$$

Design Equations: ... derivation see white paper

$$\beta_0 = b_0 = \frac{-R_4}{R_1 + R_2}$$

$$\alpha_1 = a_1 + a_2 = \frac{(R_1 \cdot (R_3 + R_4) + R_2 \cdot (R_3 + R_4) + R_3 \cdot R_4) \cdot C_3 + R_1 \cdot R_2 \cdot C_1}{R_1 + R_2} \cdot \omega_c$$

$$\alpha_2 = a_1 \cdot a_2 + a_3 = \frac{R_3 \cdot R_4 \cdot (R_1 + R_2) \cdot C_2 \cdot C_3 + R_1 \cdot (R_2 \cdot R_3 + R_2 \cdot R_4 + R_3 \cdot R_4) \cdot C_1 \cdot C_3}{R_1 + R_2} \cdot {\omega_c}^2$$

$$\alpha_3 = a_1 \cdot a_3 = \frac{R_1 \cdot R_2 \cdot R_3 \cdot R_4 \cdot C_1 \cdot C_2 \cdot C_3}{R_1 + R_2} \cdot {\omega_c}^3$$

clear $(\beta_0, \alpha_1, \alpha_2, \alpha_3, b_0, a_1, a_2, a_3, f_c, \omega_c, \omega_p, \omega_0, Q, R_1, R_2, R_3, R_4, C_1, C_2, C_3)$

Design Strategy:

- Given Values:

$A_{cl} := -1$ --> $b_0 := A_{cl}$

$f_c := 400000$ --> $\omega_c := 2 \cdot \pi \cdot f_c = 2.513 \cdot 10^6$

$a_1 := 1.0000$ $a_2 := 1.0000$ $a_3 := 1.0000$ (Butterworth)

$R_1 := 267 \cdot 10^3$ $k := 1.56$

$C_1 := 3 \cdot 10^{-12}$ $C_3 := 100 \cdot 10^{-15}$

1. Calculate R_2:

$$R_2 := k \cdot R_1 = 416.52 \cdot 10^3$$

2. Calculate R_4:

$$b_0 = \frac{-R_4}{R_1 + R_2} \quad \text{-->} \quad R_4 = -b_0 \cdot (R_1 + R_2) = -b_0 \cdot (1 + k) \cdot R_1$$

$$R_4 := -b_0 \cdot (R_1 + R_2) = 683.52 \cdot 10^3$$

3. Calculate R_3:

$$\omega_p := \frac{1}{a_1 + a_2} \cdot \omega_c = 1.257 \cdot 10^6$$

$$\omega_p = \frac{R_1 + R_2}{(R_1 \cdot (R_3 + R_4) + R_2 \cdot (R_3 + R_4) + R_3 \cdot R_4) \cdot C_3 + R_1 \cdot R_2 \cdot C_1}$$

$$R_3 := \frac{R_1 + R_2 - \omega_p \cdot (C_3 \cdot (R_1 + R_2) \cdot R_4 + C_1 \cdot R_1 \cdot R_2)}{(R_1 + R_2 + R_4) \cdot C_3 \cdot \omega_p} = 1.197 \cdot 10^6$$

$$R_3 = \frac{(1 + k) \cdot R_1 - \omega_p \cdot (C_3 \cdot ((1 + k) \cdot R_1) \cdot R_4 + k \cdot R_1^{\,2} \cdot C_1)}{((1 + k) \cdot R_1 + R_4) \cdot C_3 \cdot \omega_p}$$ (with k-Factor)

4. Conditions for C_1 and C_3:

Derived from $R_3 = \ldots$ $\quad R_1+R_2-\omega_p\cdot\left(C_3\cdot\left(R_1+R_2\right)\cdot R_4+C_1\cdot R_1\cdot R_2\right)>0$

$$C_1<\frac{R_1+R_2-\omega_p\cdot\left(R_1+R_2\right)\cdot R_4\cdot C_3}{\omega_p\cdot R_1\cdot R_2} = \frac{(1+k)+\omega_p\cdot b_0\cdot R_1\cdot(1+k)^2\cdot C_3}{\omega_p\cdot k\cdot R_1}=4.471\cdot 10^{-12}$$

$$C_3<\frac{R_1+R_2-\omega_p\cdot R_1\cdot R_2\cdot C_1}{\omega_p\cdot\left(R_1+R_2\right)\cdot R_4} = \frac{\omega_p\cdot\frac{k}{1+k}\cdot R_1\cdot C_1-1}{\omega_p\cdot b_0\cdot(1+k)\cdot R_1}=450.119\cdot 10^{-15}$$

5. Calculate C_2:

$$\omega_0:=\sqrt{\frac{a_1+a_2}{a_1\cdot a_3}}\cdot\omega_c=3.554\cdot 10^6$$

$$\omega_0=\sqrt{\frac{\left(R_1\cdot\left(R_3+R_4\right)+R_2\cdot\left(R_3+R_4\right)+R_3\cdot R_4\right)\cdot C_3+R_1\cdot R_2\cdot C_1}{R_1\cdot R_2\cdot R_3\cdot R_4\cdot C_1\cdot C_2\cdot C_3}}$$

$$C_2:=\frac{\left(R_1\cdot\left(R_3+R_4\right)+R_2\cdot\left(R_3+R_4\right)+R_3\cdot R_4\right)\cdot C_3+R_1\cdot R_2\cdot C_1}{{\omega_0}^2\cdot R_1\cdot R_2\cdot R_3\cdot R_4\cdot C_1\cdot C_3}=1.578\cdot 10^{-12}$$

6. Check b_0, ω_p, ω_0 and Q:

$R_1=267\cdot 10^3$ $\qquad C_1=3\cdot 10^{-12}$

$R_2=416.52\cdot 10^3$ $\qquad C_2=1.578\cdot 10^{-12}$

$R_3=1.197\cdot 10^6$ $\qquad C_3=100\cdot 10^{-15}$

$R_4=683.52\cdot 10^3$

$$b_0:=\frac{-R_4}{R_1+R_2}=-1$$

$$\omega_p:=\frac{R_1+R_2}{\left(R_1\cdot\left(R_3+R_4\right)+R_2\cdot\left(R_3+R_4\right)+R_3\cdot R_4\right)\cdot C_3+R_1\cdot R_2\cdot C_1}=1.257\cdot 10^6$$

--> $\quad\sim\quad \frac{1}{a_1+a_2}\cdot\omega_c=1.257\cdot 10^6$ $\qquad$ **check!**

$$\omega_0 := \sqrt{\frac{(R_1 \cdot (R_3 + R_4) + R_2 \cdot (R_3 + R_4) + R_3 \cdot R_4) \cdot C_3 + R_1 \cdot R_2 \cdot C_1}{R_1 \cdot R_2 \cdot R_3 \cdot R_4 \cdot C_1 \cdot C_2 \cdot C_3}} = 3.554 \cdot 10^6$$

--> $\sim \quad \sqrt{\frac{a_1 + a_2}{a_1 \cdot a_3}} \cdot \omega_c = 3.554 \cdot 10^6$ **check!**

$$Q := \frac{((R_1 \cdot (R_3 + R_4) + R_2 \cdot (R_3 + R_4) + R_3 \cdot R_4) \cdot C_3 + R_1 \cdot R_2 \cdot C_1) \cdot (R_1 \cdot R_2 \cdot R_3 \cdot R_4 \cdot C_1 \cdot C_2 \cdot C_3) \cdot \sqrt{\frac{(R_1 \cdot (R_3 + R_4) + R_2 \cdot (R_3 + R_4) + R_3 \cdot R_4) \cdot C_3 + R_1 \cdot R_2 \cdot C_1}{R_1 \cdot R_2 \cdot R_3 \cdot R_4 \cdot C_1 \cdot C_2 \cdot C_3}}}{((R_1 \cdot (R_3 + R_4) + R_2 \cdot (R_3 + R_4) + R_3 \cdot R_4) \cdot C_3 + R_1 \cdot R_2 \cdot C_1) \cdot (R_3 \cdot R_4 \cdot (R_1 + R_2) \cdot C_2 \cdot C_3 + R_1 \cdot (R_2 \cdot R_3 + R_2 \cdot R_4 + R_3 \cdot R_4) \cdot C_1 \cdot C_3) - R_1 \cdot R_2 \cdot R_3 \cdot R_4 \cdot (R_1 + R_2) \cdot C_1 \cdot C_2 \cdot C_3} = 0.943$$

Q must equal the Q from Tietze Schenk table. If not, adjust k below. Since these design steps are based on an approximation, we need this design knob k anyway.

--> $\sim \quad \frac{(a_1 + a_2) \cdot a_1 \cdot a_3 \cdot \sqrt{\frac{a_1 + a_2}{a_1 \cdot a_3}}}{a_1^2 \cdot a_2 + a_2 \cdot a_3 + a_1 \cdot a_2^2} = 942.809 \cdot 10^{-3}$ **check!**

7. Check α_1, α_2 and α_3:

$$\alpha_1 := \frac{(R_1 \cdot (R_3 + R_4) + R_2 \cdot (R_3 + R_4) + R_3 \cdot R_4) \cdot C_3 + R_1 \cdot R_2 \cdot C_1}{R_1 + R_2} \cdot \omega_c = 2$$

$= \quad a_1 + a_2 = 2$

$$\alpha_2 := \frac{R_3 \cdot R_4 \cdot (R_1 + R_2) \cdot C_2 \cdot C_3 + R_1 \cdot (R_2 \cdot R_3 + R_2 \cdot R_4 + R_3 \cdot R_4) \cdot C_1 \cdot C_3}{R_1 + R_2} \cdot \omega_c^2 = 2$$

$= \quad a_1 \cdot a_2 + a_3 = 2$

$$\alpha_3 := \frac{R_1 \cdot R_2 \cdot R_3 \cdot R_4 \cdot C_1 \cdot C_2 \cdot C_3}{R_1 + R_2} \cdot \omega_c^3 = 1$$

$= \quad a_1 \cdot a_3 = 1$

$$Q := \frac{\alpha_1 \cdot \alpha_3 \cdot \sqrt{\frac{\alpha_1}{\alpha_3}}}{\alpha_1 \cdot \alpha_2 - \alpha_3} = 0.943$$

8.) Fine-Tuning:

Instead of a single massive equation Q(k), use this recommended iterative design:

1. Input R_1, C_1, C_3, k and the desired filter coefficients a_1, a_2, a_3
2. Go through steps 1 to 6.
3. Adjust k:

   - If calculated Q is too **high**, decrease k.
   - If calculated Q is too **low**, increase k.

In the end, the Q in 6.) has to equal the Q from the coefficient formula in 7.). In this case, $Q = 0.943$. The calculated value of Q deviates a little from the theoretical value, since it has been approximated, as shown above. However, with the k-factor, the correct filter approximation can still be achieved. When this is the case, all the coefficients in 7.) also equal the theoretical value, coming from Tietze Schenk tables.

9.) Input Impedance:

The input impedance at dc can be approximated with $Z_{in0} := R_1 + R_2 + R_3 = 1.88 \cdot 10^6$

The input impedance transfer function will be derived soon...